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V-09—Inspection of gasoline and diesel fuel dispensers and refuellers at authorized service providers' or dealers' premises

Category: Volume

Bulletin: V-09 (rev. 4)

Document(s): W&M Act, section 26 (1); W&M Regs, section 237

Issue date: 2013-05-01

Effective date: 2013-05-01

Supersedes: V-09 (rev. 3)

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1.0 Purpose

The purpose of this bulletin is to outline the conditions under which fuel dispensers and high-speed refuellers may be initially inspected at an authorized service provider's or dealer's premises.

2.0 Scope

This bulletin applies to fuel dispensers and high-speed refuellers used to measure gasoline or diesel fuel in retail and commercial service. These are self-contained, built for purpose, measuring systems with positive displacement meters. It does not apply to coriolis, turbine or magnetic flow liquid meter technologies.

3.0 Authority

Section 26 (1) of the *Weights and Measures Act* makes it an offence for a dealer to sell, lease or otherwise dispose of a device that was not marked or inspected in the manner and circumstances prescribed.

Section 237 of the *Weights and Measures Regulations* allows meters to be tested and calibrated using a liquid other than the liquid that the meter will measure in trade.

4.0 Terminology

4.1 Authorized service provider (ASP):
is an organization that has been delegated inspection, calibration and/or certification powers pursuant to the *Weights and Measures Act*

4.2 Capillary seal:
is a seal formed by a thin layer of liquid between adjoining surfaces. The effectiveness of such a seal is partially dependant upon the clearance between surfaces and the characteristic of the fluids.

4.3 Dealer:

means any person who in the course of that person's business sells, consigns, imports, leases or lends devices.

4.4 Mechanical seal:

is a seal which through physical contact with adjoining surfaces, prevents the passage of liquid. Such seals are considered positive in nature.

5.0 Background

At the time of initial inspection, the inspector must verify the ability of the meter to accurately measure the product it is intended to measure. In the case of self-contained metering systems such as motor fuel dispensers and high speed refuellers, the physical installation of the device has little effect on the overall accuracy, thus making it metrologically possible to conduct initial inspections prior to installation. In order for such inspections to be representative of the actual conditions of use, consideration must be given to the fluid used to conduct the inspections. Gasoline is normally not an option due to safety concerns. The key factor to consider when choosing a test fluid is the design of the meter. The accuracy of meters which employ mechanical seals, typically found in motor fuel dispensers, is not significantly influenced by variations in product viscosity. Conversely, the accuracy of meters which employ capillary seals, such as high-speed refuellers and other large capacity positive displacement meters, is directly influenced by the viscosity of the fluid used to perform accuracy tests.

The relationships (size, location, capacity and adjoining plumbing) between components in metering systems, which are assembled to specifically suit a given location, directly influence the accuracy of the overall system and cannot be readily simulated in a shop or factory environment.

6.0 Policy

Given the influence of design and construction of a metering system along with the effects related to the characteristics of the fluids used during tests, the following prescribes conditions under which particular types of devices may be initially inspected at authorized service providers' or dealers' premises.

6.1 Initial inspections of metering systems may be performed at these locations provided that:

6.1.1 Test conditions applicable to the meters are as defined by Bulletin V-03, "Controlled Conditions".

6.1.2 Except for the items listed in section 6.1.4, all the accessories associated with the measuring system, as applicable, (i.e., pump, compensator, air eliminator, back pressure valve and flow limiters, etc.) are installed in the same position that they will be in after their final installation with the meter.

6.1.3 Actual flow rates are approximated.

6.1.4 A test hose and nozzle may be used in accordance with Measurement Canada and ASP test procedures during inspections of dispensers and refuellers. The substitution of other accessories and components is not permitted.

6.2 Retail motor fuel dispensers, which employ a meter with mechanical seals, intended to measure gasoline or gasoline / ethanol blends, diesel and diesel / bio-diesel blends may be inspected using test fluids with a viscosity similar to gasoline or those intended to replace diesel fuel.

6.3 High-speed refuellers which employ meters with capillary seals may be inspected using the actual product they are intended to measure or an authorized test fluid as per section 6.4.

6.4 Test fluids for the inspection of capillary seal meters intended to measure diesel fuel may be authorized for use in the following way:

6.4.1 The company or person proposing a particular liquid will provide the appropriate Regional Volumetric Specialist with relevant data, including the product temperature range at the facility and the density and viscosity of both the liquid the meter is intended to measure and the test fluid they are proposing to use. The viscosity data related to the proposed fluid must be measured by a independent, third party, laboratory. The product temperature range at the test facility must be known (or provided) to identify the optimum product viscosity. With this information, the Specialist will confirm that the characteristics of the test fluid are acceptable as a replacement for the actual diesel product. At the Specialist's discretion, subsequent follow-up inspections may be required and scheduled after the installation of specified systems.

6.4.1.1

The test fluid for diesel fuel must be a refined petroleum product or may be a combination of refined petroleum products, blended to an optimum viscosity.

6.4.1.2

The optimum viscosity for a diesel fuel test fluid is 3.9 cSt at the actual temperature of use as derived from a survey of product available in Canada. The optimum test fluid viscosity for diesel fuel can be found adjacent to the average product temperature, below.

Table 1—Optimum test fluid viscosity for a given average

product temperature

Average Product Temperature (° C)	Optimum Product Viscosity at 40° C (cSt)
5	1.9
10	2.1
15	2.3
20	2.6
25	2.9

6.4.1.3

The maximum viscosity for a diesel fuel test fluid should not exceed 4.1 cSt at 40° C.

6.4.1.4

The minimum acceptable viscosity of a test fluid for diesel fuel can be found adjacent to the maximum product temperature, below.

Table 2—Minimum test fluid viscosity for a given maximum product temperature

Maximum Product Temperature (° C)	Minimum Product Viscosity at 40° C (cSt)
10	1.7
15	1.85
20	2
25	2.25

Maximum Product Temperature (° C)	Minimum Product Viscosity at 40° C (cSt)
30	2.45
35	2.7
40	3

Example determination of optimum viscosity

A facility has determined the product flowing temperature varies from 17 to 23° C throughout the year.

The average test temperature at the facility is 20° C. From section 6.4.1.2 above, the optimum liquid should have a viscosity of 2.6 cSt at 40° C. This value is useful when the facility is planning to replace or acquire a substitute liquid. A liquid of this viscosity will provide the best calibration adjustment for the meter. Section 6.4.1.4 will provide the minimum viscosity that the test fluid must have, i.e., 2.15 cSt at 40° C. Using a liquid that is less viscous than this may result in the meter being adjusted with a significant error in favour of the seller. Liquids with higher viscosity can be used but the viscosity of the test fluid must not be greater than 4.1 cSt at 40° C.

6.4.2 Where a high-speed refueller is inspected at any location other than where it is intended to be installed, Measurement Canada inspectors and ASPs are responsible for ensuring the fluid temperature is not too high causing the viscosity to drop below the minimum values in Table 6.4.1.4. This determination can typically be done during the Automatic Temperature Compensation (ATC) test. The temperature will be recorded as an annotation in the comment section of the inspection certificate.

6.4.3 It is the responsibility of the ASP for maintaining the fluid viscosity within acceptable limits and retaining objective evidence of such, from a third party independent laboratory. Such testing should be done each time the fluid is replaced or every two years.

7.0 Revision

The original bulletin was issued on March 31, 1995, as Bulletin V-12. It documented, in detail, three stages of a survey which was conducted in conjunction with the Canadian Gas Pump Manufacturers Association. The bulletin authorized the setting of a target calibration bias.

7.1 The purpose of Revision 1 (Bulletin V-12) was to remove the setting of the target calibration bias.

7.1.1 Revision 1 was reformatted and renumbered to Bulletin V-9 (rev.1) in a revised set of bulletins and issued on March 31, 1995.

7.2 The purpose of Revision 2 was to revise the bulletin format and sequence; includes minor editorial modifications for clarification and identifies new references to associated bulletins; updated new agency name.

7.3 The purpose of Revision 3 was to remove information which was no longer relevant and to outline conditions for the inspection of gasoline and diesel fuel dispensers and to include refuellers which may be initially inspected at authorized service providers' or dealers' premises; and to:

a. define the difference between mechanical (positive) sealed meters (i.e., dispensers) and capillary sealed meters (i.e., high-speed refuellers); and
b. establish the process for authorizing new test fluids when they are proposed.

7.4 The purpose of Revision 4 in revised section 6.1.2 and 6.1.4 is to allow the use of a test hose and nozzle during factory inspections of fuel dispensers and refuellers.

Date modified:
2017-03-28